Herald

Herald — Messenger Channels (Operator Guide)

Field	Value
Revision	3
Created	2026-05-28
Last modified	2026-05-31
Status	active
Status summary	r3: added §6B (Intent recognition — subscribers speak plain natural language) documenting the intent-recognition contract (`docs/design/INTENT_RECOGNITION.md`) — NO command syntax / no `COMMAND:` prefix; the three-tier resolution (Tier 1 deterministic CommandRecognizer fast-path → Tier 2 LLM intent inference → Tier 3 `clarify` reply-tag-and-ask fallback); the command set the System recognizes; and the never-guess / never-ignore behaviour. Prior r2: added §6A (Participant identity, attribution & @-tagging) documenting the participant/attribution contract (`docs/design/PARTICIPANT_ATTRIBUTION.md`) — per-channel `subscribers/subscriber_aliases.username` mapping, the `HERALD_<CHANNEL>_OPERATOR_USERNAME` operator env var (`HERALD_TGRAM_OPERATOR_USERNAME=@milos85vasic`), and the @-tagging matrix (Claude + Operator never tagged). Operator-facing reference for Herald's Wave 7 generic messenger-channel framework. Documents the `commons_messaging/channels.Channel` interface (the richer inbound contract that embeds the §11.0 outbound `commons.Channel`), the `init()`-driven registry that resolves channel names at runtime, the per-channel content-addressed inbox (`~/.herald/inbox/<channel>/<sha256>.<ext>`), the generalized self-filter that defends every adapter against echo-loops, the `HERALD_CHANNELS` selector that drives multi-channel `pherald listen`, the two live adapters (Telegram since Wave 6, Slack since Wave 7 T6), the URL-scheme parsers, the implementer checklist for adding a new channel, the troubleshooting cookbook, and the operator pre-deploy audit checklist.
Issues	none
Issues summary	—
Fixed	(n/a — new guide)

Field	Value
Continuation	bump when new channels land (Wave 8 Max/Email per `docs/CONTINUATION.md` §3); bump when the channels registry gains hot-reload or per-channel rate-limit knobs; bump when a `pherald listen --dry-run` (registry-resolution probe) ships.

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§1. Architecture overview

§1.1 Two interfaces, one contract

Herald's messenger plane has two complementary contracts. The outbound contract is the **spec §11.0** `commons.Channel` — the five-method shape every flavor's `serve` plane depends on (`Name`, `Capabilities`, `Send`, `Subscribe`, `HealthCheck`). Widening that interface would ripple into every flavor's outbound path, so Wave 7 introduced a strictly richer inbound contract in a NEW package without touching the existing one:

Layer	Package	Methods	Consumers
Outbound contract (§11.0)	`commons`	`Name`, `Capabilities`, `Send`, `Subscribe`, `HealthCheck`	every flavor's `serve` plane
Inbound contract (Wave 7)	`commons_messaging/channels`	the five above (embedded) plus `SendReplyGeneric`, `BotSelfIdentity`, `DownloadAttachment`	`pherald listen`, `qaherald`

The richer interface lives at `commons_messaging/channels/channel.go`. Every concrete adapter (Telegram, Slack, future Max/Email/Discord) satisfies it; the inbound runtime depends ONLY on the interface, never on the concrete adapter type, so adding a new channel does NOT touch `pherald listen`'s hot path.

§1.2 The eight-method shape (full)

```
type Channel interface {
    commons.Channel                                          // 5
        outbound methods (§11.0)
    SendReplyGeneric(                                        //
        quoted reply with attachments
        ctx context.Context,
        recipient commons.Recipient,
        body string,
        replyToID string,
        attachments []commons.Attachment,
    ) (string, error)
    BotSelfIdentity(ctx context.Context) (SelfIdentity,
        error)  // anti-echo-loop
    DownloadAttachment(                                       //
        content-addressed inbox write
        ctx context.Context,
        externalID string,
        mime string,
    ) (string, string, error)                                 //
        (finalPath, sha256Hex, err)
}
```

A `SelfIdentity` is a `{Kind, Value}` pair where `Kind` is one of `IdentityUsername` (Telegram `bot.Me.Username`), `IdentityUserID` (Slack `bot_user_id`), or `IdentityAddress` (Email From-address — Wave 8 reserved).

§1.3 Reply-method naming rationale (load-bearing footnote)

`SendReplyGeneric` exists rather than `SendReply` because `*tgram.Adapter` predates the interface and already exposes a native `SendReply(ctx, chatID int64, replyToMessageID int, ...)`. A Go type may hold only one method named `SendReply`; the only way to preserve the existing tgram behaviour while adding the channel-agnostic method was to give the new method a different name. The matching `inbound.Replier` interface (`pherald/internal/inbound`) uses the shorter `SendReply` because it is freshly introduced — and is bridged to `Channel.SendReplyGeneric` by the thin `channelReplier` shim in `pherald/cmd/pherald/listen.go`.

§1.4 The registry — name → constructor

`commons_messaging/channels/registry.go` is a tiny package-global `map[string]Constructor` populated by each adapter's `init()`. The registry is the entire abstraction between the adapter implementation and the runtime: `pherald listen` blank-imports the adapter packages, the adapters self-register, and `channels.New("tgram", cfg)` (or `"slack"`, etc.) resolves by name.

```
// commons_messaging/channels/tgram/tgram.go
func init() {
```

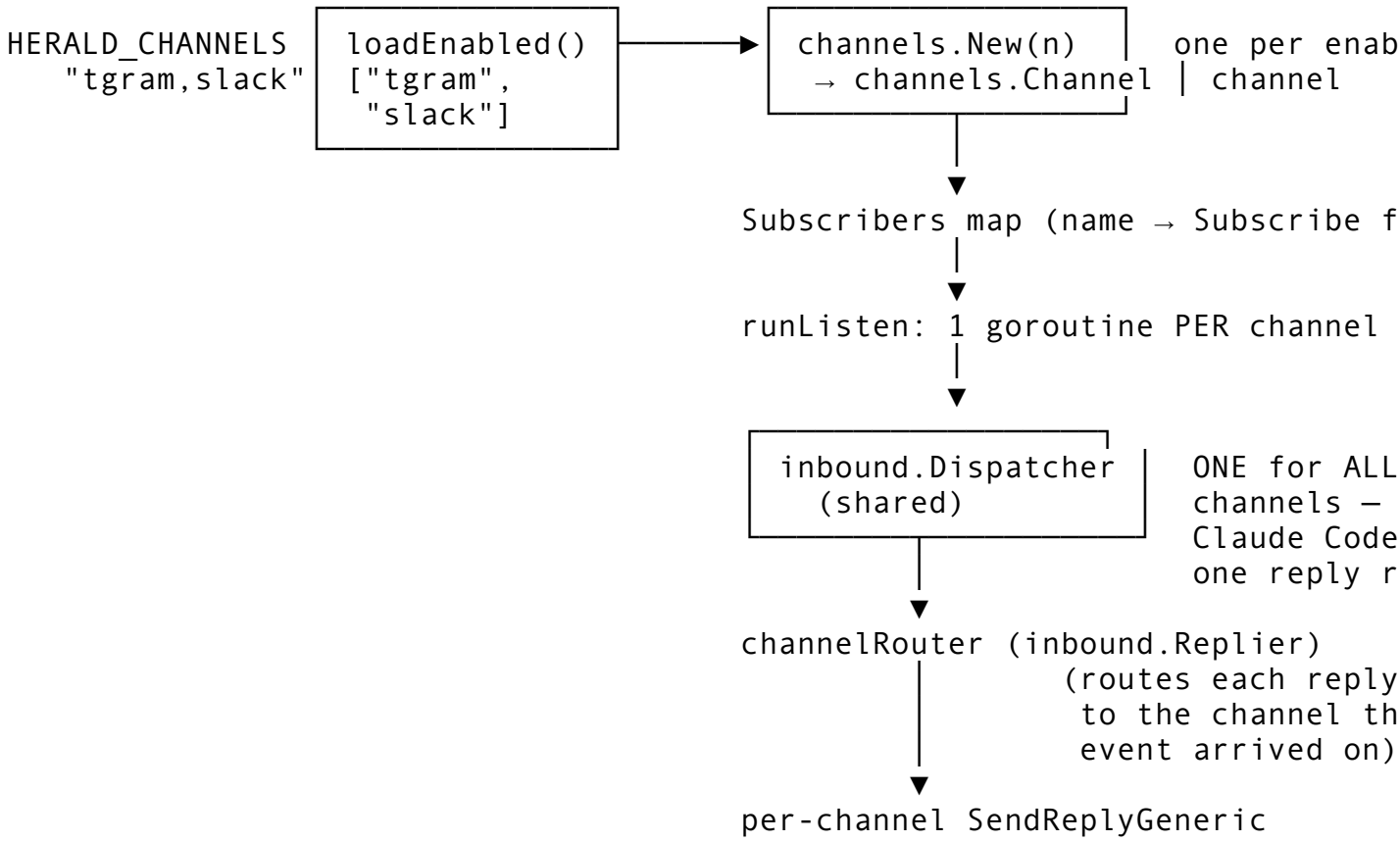
```
    channels.Register(string(commons.ChannelTelegram), func(cfg
        channels.Config) (channels.Channel, error) {
        if cfg.BaseURL != "" {
            return NewAdapterWithBaseURL(cfg.Token, cfg.Target,
        cfg.BaseURL), nil
        }
        return NewWithCreds(cfg.Token, cfg.Target), nil
    })
}
```

Duplicate registration **panics** at process start — a typo that shadowed one of two adapters with the same name would silently route traffic to the wrong adapter (a classic §107 PASS-bluff), so the registry refuses to load.

`channels.Names()` returns the sorted list of registered names; `channels.New(unknown, cfg)` returns `ErrUnknownChannel` with the registered set embedded in the error message (fail-loud — no silent no-op channel).

§1.5 The runtime — `pherald listen` fan-in

`pherald listen` (`pherald/cmd/pherald/listen.go`) wires the registry to the inbound dispatcher:



The fan-in semantics are deliberate: ONE Claude Code session sees subscriber traffic from EVERY enabled channel, and the dispatcher's `channelRouter` (a small `map[string]inbound.Replier` keyed by channel name) routes each reply back to the adapter the inbound event arrived on. A Slack message gets a Slack reply; a Telegram message gets a Telegram reply; there is no cross-channel reply leak. If any subscriber goroutine returns a

non-cancellation error, `runListen` cancels its siblings and propagates the error — silent partial failure is forbidden (`pherald/cmd/pherald/listen.go:399..462`).

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§2. Enabling channels (`HERALD_CHANNELS`)

`HERALD_CHANNELS` is the single env var that controls which channels Wave 7 `pherald listen` brings up:

Value	Behaviour
unset or empty	Wave 6 single-channel default — Telegram only (`tgram`).
`tgram`	Telegram only. Identical to the default.
`tgram,slack`	Both Telegram and Slack — one Subscribe goroutine per channel, one shared inbound dispatcher.
`slack`	Slack only (Telegram disabled — no `HERALD_TGRAM_*` required).
`tgram,foo`	**FAILS LOUD AT BOOT** — `foo` is not registered, so `channels.New("foo")` returns `ErrUnknownChannel`.

Whitespace inside the value is tolerated; ordering does not matter; duplicates are collapsed; leading/trailing/double commas are skipped. The parser is `loadEnabledChannels()` (`pherald/cmd/pherald/listen.go:289..305`).

Fail-loud invariants.

1. Unknown channel name → `pherald listen` refuses to boot with `ErrUnknownChannel: "foo" (registered: [slack tgram])`.
2. An enabled channel whose required env is unset → boot fails with a per-channel descriptive error (e.g. `HERALD_SLACK_BOT_TOKEN required (channel slack enabled)`).
3. Empty resolved channel set (a refactor regression) → boot fails with `no channels enabled (HERALD_CHANNELS resolved empty)`.
4. Any subscriber goroutine that returns a non-cancellation error after boot → `runListen` cancels every sibling subscriber and returns the error to the caller. Silent partial degradation is impossible by construction.

No `--allow-unknown-channels` / `--skip-credential-check` / `--single-channel-fallback` escape hatch exists; per §107 anti-bluff, a silently-degraded listener is worse than a hard failure because it claims to work for end users.

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§3. Telegram (LIVE — Wave 6 HRD-011)

§3.1 Required environment

Variable	Required for	Format	Example
`HERALD_TGRAM_BOT_TOKEN`	outbound + inbound	`<bot-id>:<api-token>` from BotFather	`1234567890:AAH1lab2c3d4e5f6g7h8i9j`

Variable	Required for	Format	Example
HERALD_TGRAM_CHAT_ID	outbound default destination	numeric (group: negative, supergroup: starts -100, DM: positive)	-1001234567890

§3.2 Obtaining a bot token

1. Open Telegram and DM @BotFather.
2. Send /newbot. Pick a display name (any string) and a username ending in bot (must be globally unique).
3. BotFather replies with Use this token to access the HTTP API: followed by your token. Copy it into HERALD_TGRAM_BOT_TOKEN.
4. (Optional — strongly recommended for groups.) Send /setprivacy to BotFather, select your bot, choose Disable. Without this, group members' messages are NOT delivered to your bot via getUpdates. Privacy-mode-on is the single most common cause of "the bot doesn't see my messages" tickets.

§3.3 Obtaining a chat id

For a 1:1 DM:

1. Send any message to your bot.
2. Curl https://api.telegram.org/bot<TOKEN>/getUpdates and read the result[].message.chat.id field — that integer is HERALD_TGRAM_CHAT_ID.

For a group / supergroup:

1. Add the bot to the group, send a message.
2. Curl getUpdates as above. Group chat ids are negative integers; supergroup chat ids start with -100.

§3.4 Transport (under the hood)

- **Outbound.** sendMessage (text), sendPhoto / sendDocument / sendVoice (attachments). Markdown rendering uses Telegram's MarkdownV2 dialect.
- **Inbound.** getUpdates long-poll with timeout=25s. Each poll cycle drains all pending updates, hands them to OnPhoto / OnDocument / OnVoice / generic message handlers, and re-polls.
- **Self-filter.** bot.Me.Username is read once at Subscribe construction (cached for the process lifetime) — see commons_messaging/channels/tgram/subscribe.go.
- **Reply threading.** reply_to_message_id quotes the originating message. Forum-topic threading via message_thread_id is supported in V2.
- **Attachments.** Photos / documents / voice messages are streamed via getFile + the file-download endpoint into ~/.herald/inbox/tgram/<sha256>.<ext> (see §5).

§3.5 Capabilities

```
Text                : true
Markdown            : true   (MarkdownV2 dialect)
HTML                : true
Attachments         : true
AttachmentMaxMiB    : 50    (Telegram Bot API hard limit)
Threads             : true  (forum topics via message_thread_id)
InteractiveURL      : true
InteractiveCall     : false (V2 advanced — HRD-011 follow-up)
DeliveryCeiling     : DeliveryRouted (200 OK == platform stored)
```

§3.6 Reading the diary

Every message in and out is appended to `docs/herald/diary/main.md` (and re-exported to `main.pdf` / `main.html` per the conversation-diary invariant — see `CLAUDE.md`).

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§4. Slack (LIVE — Wave 7 T6 HRD-115)

§4.1 Required environment

Variable	Required for	Format	Example
`HERALD_SLACK_BOT_TOKEN`	outbound + identity	starts `xoxb-`	`xoxb-1234-5678-AbCdEfGh`
`HERALD_SLACK_APP_TOKEN`	inbound (Socket Mode)	starts `xapp-`	`xapp-1-A0123-456-XYZ`
`HERALD_SLACK_CHANNEL_ID`	outbound default destination	starts `C` (public) / `G` (private)	`C0123456789`

`HERALD_SLACK_APP_TOKEN` is required ONLY when the Slack channel participates in inbound (i.e. when `HERALD_CHANNELS` includes `slack`). An outbound-only Slack deployment can omit it.

§4.2 Creating a Slack app

1. Visit https://api.slack.com/apps and click **Create New App → From scratch**.
2. Name the app, pick the target workspace, click **Create**.
3. In the app's settings, navigate to **OAuth & Permissions → Scopes → Bot Token Scopes** and add:
   - `chat:write` — required for `chat.postMessage`.
   - `files:read` — required for inbound attachment retrieval.
   - `files:write` — required for outbound attachment upload (`files.uploadV2`).
   - `channels:history` — required for public-channel inbound. Use `groups:history` instead if the bot will live in private channels.
   - `users:read` — recommended, lets Herald enrich the sender display name in the diary.
4. Navigate to **Socket Mode** and toggle **Enable Socket Mode**. Slack then prompts you to generate an app-level token with the `connections:write` scope — generate it and copy the `xapp-…` value into `HERALD_SLACK_APP_TOKEN`.

5. Navigate to **Event Subscriptions**, toggle **Enable Events**, then subscribe the bot to (at minimum) `message.channels` (public) and/or `message.groups` (private). Socket Mode means there is NO public webhook URL to configure — events arrive over the WebSocket.
6. Navigate to **Install App → Install to Workspace**. Authorize the install; Slack returns the **Bot User OAuth Token** (`xoxb-…`) — copy it into `HERALD_SLACK_BOT_TOKEN`.
7. Add the bot to the target channel: in the Slack client run `/invite @your-bot-name`. Then right-click the channel name → **View channel details** → copy the channel id (`C…` or `G…`) into `HERALD_SLACK_CHANNEL_ID`.

§4.3 Transport (under the hood)

- **Outbound.** `chat.postMessage` for text and Slack-flavored markdown (`mrkdwn`); `files.uploadV2` for attachments.
- **Inbound.** Socket Mode — a persistent WebSocket originated by Herald to Slack's edge. No inbound NAT traversal, no public URL, no webhook signature verification — Slack's onus, not the operator's. This is the equivalent of Telegram's `getUpdates` long-poll.
- **Self-filter.** `auth.test` is called once at Subscribe construction. The returned `bot_user_id` (a `U…` value such as `U01ABC`) is cached for the process lifetime; an empty result is fail-loud (see §6).
- **Reply threading.** Replies use `thread_ts` — Slack's anchor for reply threading. The `channelRouter` passes the originating message's `ts` through to `SendReplyGeneric` as `replyToID`, the Slack adapter rewrites it as `thread_ts`.
- **Attachments.** Inbound attachments arrive as `file_share` subtype events; the adapter follows `file.url_private_download` (presigned, requires the bot token in the `Authorization: Bearer …` header) and streams the bytes into `~/.herald/inbox/slack/<sha256>.<ext>` (see §5).

§4.4 Capabilities

```
Text                : true
Markdown            : true   (Slack mrkdwn dialect — *bold*, _italic_, `co
HTML                : false  (Slack rejects HTML — adapters render mrkdwn
Attachments         : true
AttachmentMaxMiB    : 1024   (workspace-plan dependent; this is the API so
Threads             : true   (thread_ts)
InteractiveURL      : true
InteractiveCall     : false  (interactive Block Kit / call routing — futur
DeliveryCeiling     : DeliveryRouted (postMessage 200 == platform stored)
```

§4.5 §107 security note — token redaction

`xoxb-…` tokens are workspace-scope credentials that, if leaked into a log line or error message, give the holder full bot privileges. The Slack adapter and the `qaherald` Slack scenario BOTH carry a `sanitizeError` step that scrubs `xoxb-` / `xapp-` substrings from every error returned to the caller. The invariant is pinned by `TestSlack_Send_ErrorDoesNotLeakToken` (`commons_messaging/channels/slack/send_test.go`) — a mutation that disabled the sanitizer would surface immediately. Operators should still avoid `set -x` in production wrappers and should rotate any token that appears in a captured trace.

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§5. Per-channel inbox isolation

Every inbound attachment is content-addressed: the sha256 of the bytes IS the filename stem; the MIME-derived extension is the only suffix. The directory is namespaced by channel so a Telegram photo and a Slack photo with identical bytes do NOT collide and the on-disk forensic trail is self-describing:

```
~/.herald/inbox/
├── tgram/
│   ├── 6a8d2c1f….jpg     (Telegram photo)
│   └── 9b3e4d5f….ogg     (Telegram voice)
├── slack/
│   ├── 6a8d2c1f….jpg     (Slack file with byte-identical content — coexis
│   └── 7c8a9b0d….pdf     (Slack document)
└── …
```

Override the inbox root via `HERALD_INBOX_DIR` (the parent of the per-channel subdirs); the per-channel subdir naming is fixed and matches the channel's registered name. Permissions are `0700` on the per-channel directory (no other-user read).

The write helper is `channels.WriteContentAddressed(channel, mime, body)` (`commons_messaging/channels/inbox.go`). It uses `io.MultiWriter(tempFile, sha256Hasher)` to hash inline as bytes stream to disk — no full-payload buffering — then atomic-renames into the final content-addressed path. Idempotence is built in: if the content-addressed final path already exists (a duplicate download), the temp file is discarded and the existing path is returned unchanged. The corresponding tests (`inbox_test.go` + adapter-level `attachments_test.go`) pin the exact path shape, byte-equality, and absence of `.part` residue.

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§6. Self-filter (anti-echo-loop)

Every Herald channel adapter is a bot that BOTH sends and receives in the same conversation. Without active filtering, every reply the dispatcher emits would arrive back as a fresh subscriber message, triggering another Claude Code dispatch, triggering another reply — an exponential loop that would saturate the long-poll within the 25-second timeout on the first reply. Wave 6 defended this in `tgram` via `bot.Me.Username`; Wave 7 generalized the defence to a channel-agnostic `BotSelfIdentity` + `IsSelfEcho` pair.

§6.1 The flow

1. **Adapter construction time:** `Subscribe` is called. Before returning, the adapter calls its own `BotSelfIdentity(ctx)` (the channel's native identity API — Telegram's `getMe`, Slack's `auth.test`, future Email's From-address resolver). The result is cached.
2. **Boot fail-loud:** if the identity result is empty (`SelfIdentity.Value == ""`), `Subscribe` refuses to start. An echo-defenceless listener is forbidden by construction — there is no `--allow-empty-identity` knob.
3. **Per-message:** the adapter's inbound handler builds an `InboundEvent` and calls `channels.StampSender(ev.Raw, isBot, kind, value)` to stamp the sender's native identity into the event's raw map.
4. **Filter time:** `channels.IsSelfEcho(ev, self)` compares the stamped sender identity against the cached self-identity — same `Kind`, same `Value`, sender flagged as bot → echo, drop. Different bot (or human user) → keep, dispatch to Claude Code.

§6.2 The comparator

`IsSelfEcho` (`commons_messaging/channels/selffilter.go`) is deliberately narrow:

- Different-bot messages are KEPT — multi-bot collaboration in the same conversation is real subscriber traffic and must not be silently dropped.
- Empty self-identity is treated as a conservative KEEP — but Subscribe should have already refused to boot in that case, so reaching the filter with empty self is a defence-in-depth fallback that surfaces as duplicate traffic (visible to the operator) rather than silent message loss.
- Human-user messages are KEPT — `RawSenderIsBot == false` short-circuits to KEEP without even comparing identities.

§6.3 Per-channel identity kinds

Adapter	`IdentityKind`	**Source**	**Example** `Value`
Telegram	`IdentityUsername`	`getMe` → `result.username`	`herald_bot` (no @)
Slack	`IdentityUserID`	`auth.test` → `user_id`	`U01ABCDEF`
Email (Wave 8)	`IdentityAddress`	configured From-address	`herald@example.com`

A new channel chooses the `IdentityKind` that the channel's native event API exposes on every inbound event; if the channel has no native sender identity (a truly anonymous channel), the channel cannot host bidirectional bot traffic at all — that constraint is upstream of Herald.

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§6A. Participant identity, attribution & @-tagging (per `docs/design/PARTICIPANT_ATTRIBUTION.md`)

Beyond the self-filter's bot-identity check (§6, which only answers "is this message from MY own bot?"), Herald relates every message to a **Participant** — a logical Subscriber/User. The same logical person may have a DIFFERENT username on every messenger, so identity is resolved per channel. This is the contract in [`docs/design/PARTICIPANT_ATTRIBUTION.md`], inherited from HelixConstitution per §11.4.35.

§6A.1 Per-channel username mapping

A Participant is backed by two PG tables:

- `subscribers` — the logical party: canonical messenger-neutral `handle`, `display_name`, `kind ∈ {human, agent, service}`.
- `subscriber_aliases` — the per-channel handle: `subscriber_id`, `channel`, `channel_user_id`, + **NEW** `username TEXT` (the per-channel `@handle` used for @-tagging — distinct from `channel_user_id`, which is the chat/user id). `UNIQUE (channel, channel_user_id)`.

The **canonical handle** (the messenger-neutral string stored in a workable item's `created_by`/`assigned_to`) is either `Claude` (the reserved system-agent sentinel — `kind=agent`,

NEVER tagged) or a human's canonical handle (defaults to their Telegram `@username` since Telegram is the primary messenger). Resolution bridges the two worlds:

- inbound — `ResolveSender(channel, channel_user_id, username)` maps a received message's sender to a canonical handle;
- outbound — `UsernameFor(handle, channel)` returns the `@username` for a canonical handle on a target channel, reporting not-found when the participant has no alias there (you cannot tag someone who is not on that messenger).

§6A.2 Operator env var

The **operator** — the one human who drives the system via the Claude Code CLI — is designated by env var, NOT a DB flag:

Env var	Example value	Meaning
`HERALD_TGRAM_OPERATOR_USERNAME`	`@milos85vasic`	the operato Telegram `@userna` (primary messenger
`HERALD_<CHANNEL>_OPERATOR_USERNAME`	`HERALD_SLACK_OPERATOR_USERNAME=…`	per-messenger generaliza for any oth channel

The operator's canonical handle equals this env value (`IdentityResolver.OperatorHandle()`).

§6A.3 @-tagging behaviour

On any workable-item event, the outbound notification to each channel @-tags the participant(s) who must be aware, resolved to that channel's `@username`:

- tag `assigned_to` when it is a human handle AND `assigned_to != Operator`;
- tag `created_by` when it is a human handle AND `created_by != Operator` AND `created_by != "Claude"`;
- `"Claude"` is NEVER tagged (it is the system); the Operator is NEVER tagged (no self-ping);
- de-dup, then resolve each handle to the channel's `@username` and skip any participant with no alias on that channel.

The `tgram` adapter renders a mention as `@username` (a Telegram username mention reaches a group member); other adapters render their channel's native mention syntax (Slack `<@U…>`, etc. — future). Attribution columns + the wiring points are documented in `WORKABLE_ITEMS_INTEGRATION.md` §3.6–§3.8.

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§6B. Intent recognition (subscribers speak plain natural language, per `docs/design/INTENT_RECOGNITION.md`)

Subscribers on **every** channel speak **plain natural language**. There is **no command syntax** — no `COMMAND:` prefix, no fixed grammar, nothing to memorize. A subscriber writes a clear message in their own words and the System determines the intent. "can you close ATM-123 please" is exactly as valid as any other phrasing. The single authoritative contract is [`docs/design/INTENT_RECOGNITION.md`](docs/design/INTENT_RECOGNITION.md).

§6B.1 Three-tier resolution

Every inbound subscriber message (from any channel) is resolved to exactly one action via three tiers, in order — the first that succeeds wins:

1. **Tier 1 — command recognition.** A deterministic `CommandRecognizer` (`pherald/internal/inbound`) maps a clear natural-language imperative to a structured action WITHOUT an LLM round-trip (fast-path; no prefix). It is conservative — only a confident match fast-paths, otherwise it defers to Tier 2.
2. **Tier 2 — intent inference.** When no command matches, the shared Claude Code dispatch (the LLM) infers the intent from the message and returns the action. The dispatch envelope instructs the LLM to recognize Herald's command set, map natural language to the right action, and never guess.
3. **Tier 3 — clarify (fallback).** When neither a command nor a confident intent can be determined, the System **replies** to the original message, **@-tags the sender** (resolved via the §6A participant `@username` mapping), and asks a **precise** clarifying question naming the candidate intents (e.g. "@user, did you want to close ATM-9, reassign it, or just get its status?").

§6B.2 The command set the System recognizes

Recognized intents (case-insensitive, phrasing-tolerant — examples, not an exhaustive grammar):

Natural-language intent (examples)	resulting action
"close ATM-123", "mark ATM-5 fixed/done/resolved"	update the item's status
"set ATM-9 to in progress", "ATM-9 is blocked"	update the item's status
"assign ATM-5 to @bob", "give ATM-5 to @bob"	reassign the item
"open a bug: …", "create a task: …", "new feature request: …"	open a new item (created_by = sender)
"investigate ATM-7", "look into ATM-7"	start an investigation
"status of ATM-9?", "what's ATM-9?"	reply with the item's status
any question / conversational message	a conversational reply
ambiguous / unparseable	a clarifying question (Tier 3)

§6B.3 Never guess, never ignore

Two rules bind every channel: a **wrong** action is worse than a clarifying question, so the System **never guesses** an action; and every inbound message resolves to exactly one action, so a subscriber is **never ignored** — genuine ambiguity is always answered with a tagged, threaded clarifying reply. This is the anti-annoyance guarantee: the subscriber never has to learn syntax and always gets a response.

§7. URL schemes

Adapters accept both env-driven `channels.Config` construction (via the registry) AND a URL-form for ad-hoc / test usage. The URL is parsed at construction so the env-driven path goes through the same parser as the test path.

§7.1 Telegram

`tgram://<bot-token>@<bot-username>/<chat-id>`

- `<bot-token>` — required, the `1234:AAH…` token (URL-encoded `:` is fine).
- `<bot-username>` — informational; carried to the diary annotation. The actual identity is re-read from `getMe` at Subscribe time, so this value is NOT trusted for security decisions.
- `<chat-id>` — required, the numeric default destination.

Example: `tgram://1234567890%3AAAH1lab%2D…@herald_bot/-1001234567890`.

§7.2 Slack

`slack://<bot-token>@<workspace>/<channel-id>[?app_token=<xapp-…>]`

- `<bot-token>` — required, the `xoxb-…` token.
- `<workspace>` — informational (mirrors the per-host convention so op-supplied config can carry it for diary annotation); NOT required to dial the Slack API.
- `<channel-id>` — required, the `C…` or `G…` channel id.
- `app_token` — required for Socket Mode (Subscribe). Optional for an outbound-only deployment.

Example: `slack://xoxb-1234-5678@acme/C0123456789?app_token=xapp-1-A0123-456-XYZ`.

§7.3 Where the env path lives

`pherald listen` does NOT consult URLs from env directly. It reads namespaced env vars (`HERALD_TGRAM_*`, `HERALD_SLACK_*`) into a `channels.Config` and constructs via the registry. The URL form is used by `qaherald` scenario configs and by hermetic tests that want a single-string fixture.

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§8. Adding a new channel

The framework's hot paths (registry resolution, multi-channel fan-in, generalized self-filter, per-channel inbox, reply routing, qaherald scenario harness) all work for free once a new adapter satisfies `channels.Channel`. The implementer checklist:

§8.1 Implementer checklist

- [ ]

- [ ] **1. Create the package.** New directory `commons_messaging/channels/<yourname>/` with a `<yourname>.go` containing the `Adapter` type. Reference the Slack adapter as a working template (`commons_messaging/channels/slack/`) — it is the post-Wave-7 reference implementation and intentionally smaller than Telegram's substrate.

- [ ] **2. Satisfy the eight methods.** Implement `Name`, `Capabilities`, `Send`, `Subscribe`, `HealthCheck` (the §11.0 outbound contract) plus `SendReplyGeneric`, `BotSelfIdentity`, `DownloadAttachment`. A compile-time assertion `var _ channels.Channel = (*Adapter)(nil)` at file scope catches signature drift on the first build.

- [ ] **3. Register at init().** Add `func init() { channels.Register("yourname", func(cfg channels.Config) (channels.Channel, error) { … }) }`. Validate required `cfg` fields in the constructor and return an explicit error on missing creds — never construct an adapter that will fail at first wire crossing (a §107 PASS-bluff).

- [ ] **4. Blank-import.** Add `_ "github.com/vasic-digital/herald/commons_messaging/channels/yourname"` to `pherald/cmd/pherald/listen.go` and (if you want qaherald coverage) to `qaherald/cmd/qaherald/main.go`. The blank import forces `init()` to run at process start so the registry resolves your channel by name.

- [ ] **5. Namespace your env.** Pick a stable short token (`max`, `email`, `discord`) and use it as the prefix for every credential: `HERALD_<YOURNAME>_BOT_TOKEN`, `HERALD_<YOURNAME>_CHANNEL_ID`, etc. Add a branch to `perChannelConfig()` in `pherald/cmd/pherald/listen.go` so unknown-env errors are channel-named (e.g. `HERALD_DISCORD_BOT_TOKEN required (channel discord enabled)`).

- [ ] **6. Inbox subdir.** No code needed — `channels.WriteContentAddressed(a.Name(), …)` already picks `~/.herald/inbox/<yourname>/`. Just confirm your `Name()` returns the same string you `Register()`ed.

- [ ] **7. Self-filter wiring.** In your Subscribe handler, call `channels.StampSender(ev.Raw, isBot, kind, value)` on every event you build; use the matching `IdentityKind` for your channel's native identity. The filter logic in `inbound.Dispatcher` then drops self-echoes for free.

- [ ] **8. URL parser.** Implement `NewFromURL("yourname://…")` for parity with Telegram/Slack. Round-trip test it.

- [ ] **9. Tests.** Hermetic tests using `httptest.NewServer` to mock the channel API (see `slack/*_test.go` and `tgram/*_test.go` for the pattern). At minimum: `TestSend_…`, `TestSendReply_…`, `TestSubscribe_…`, `TestBotSelfIdentity_…`, `TestDownloadAttachment_…`, `TestNewFromURL_…`. Pin token redaction if your channel uses sensitive credentials.

- [ ] **10. qaherald scenario.** Add a `MessengerClient` implementation under `qaherald/internal/messenger/<yourname>/` and a `builder` case so `qaherald` can

drive round-trip QA against your channel. The bidirectional transcript lands under `docs/qa/HRD-NNN-<run-id>/`.

- [ ] **11. e2e + mutation gates.** Add positive-evidence invariants to `scripts/e2e_bluff_hunt.sh` and (when a release cycle next runs a mutation gate) paired mutations covering the critical paths (`Send`, `SendReplyGeneric`, `BotSelfIdentity`).

- [ ] **12. Docs.** Add a `§X. <Channel>` section to THIS guide between Telegram (§3) and the troubleshooting section (§9), following the §3/§4 template (required env, obtaining credentials, transport details, capabilities table, security notes).

- [ ] **13. Spec V3 row.** Add the channel to `docs/specs/mvp/specification.V4.md` §11 capabilities matrix and §43 catalogue.

- [ ] **14. Issues row.** Open the implementing HRD in `docs/Issues.md`; migrate to `docs/Fixed.md` atomically on close per §11.4.19.

§8.2 What you do NOT touch

The following are channel-agnostic and stay untouched when a new channel lands:

- `pherald/cmd/pherald/listen.go` core dispatcher wiring (the only addition is your blank-import + `perChannelConfig` branch).
- `pherald/internal/inbound/dispatcher.go` (the action-routing core).
- `commons_messaging/channels/channel.go` (the interface).
- `commons_messaging/channels/registry.go` (the registry).
- `commons_messaging/channels/selffilter.go` (the comparator).
- `commons_messaging/channels/inbox.go` (the content-addressed write helper).

A working Wave 7 reference for the "minimal new channel" case is the Slack adapter (`commons_messaging/channels/slack/`) — ~600 lines including tests, all conforming to the checklist above.

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§9. Troubleshooting

§9.1 Echo loop after the first few replies

Symptom. The bot replies once; then a second, third, fourth reply arrives within seconds; the long-poll never quiesces.

Root cause. `BotSelfIdentity` returned an empty `Value` (or the wrong identity), so `IsSelfEcho` does not match the bot's own messages and they re-trigger Claude Code.

Fix.

1. Confirm the adapter's Subscribe constructor refused to boot on empty identity — if it did boot, the fail-loud invariant was broken; file a bug.
2. Check the identity API result manually:
   - Telegram: `curl -s "https://api.telegram.org/bot<TOKEN>/getMe" | jq '.result.username'` — must be non-empty.

    - Slack: `curl -s -H "Authorization: Bearer <xoxb-…>" https://slack.com/api/auth.test | jq '.user_id'` — must start with `U`.
3. Confirm the adapter's `StampSender` call carries the same `IdentityKind` the bot self-identity reports (a Telegram adapter that stamps `IdentityUserID` instead of `IdentityUsername` would not match).

§9.2 Slack `channel_not_found`

Symptom. `chat.postMessage` returns `{"ok":false,"error":"channel_not_found"}`.

Root cause.

1. `HERALD_SLACK_CHANNEL_ID` is a channel name (`#general`) rather than the channel id (`C…`/`G…`). Names are NOT accepted by the Web API.
2. The bot has not been invited to the channel — run `/invite @your-bot-name` in the target channel from a workspace member account.
3. The bot lacks the right history scope — `channels:history` for public, `groups:history` for private. Re-install the app after adding scopes.

§9.3 Slack `invalid_auth` or `token_expired`

Symptom. Every Web API call returns `{"ok":false,"error":"invalid_auth"}` or `token_expired`.

Root cause. The `xoxb-…` token has been revoked, the workspace's owner rotated app credentials, or the app was uninstalled and reinstalled (re-installation issues a new token).

Fix. Rotate `HERALD_SLACK_BOT_TOKEN` to the freshly-issued value from **OAuth & Permissions → Bot User OAuth Token**. Restart `pherald listen`.

§9.4 Slack `not_allowed_token_type` for `apps.connections.open`

Symptom. Socket Mode WebSocket dial fails immediately.

Root cause. `HERALD_SLACK_APP_TOKEN` is empty, or carries the bot token by mistake. Socket Mode connect requires the `xapp-…` app-level token, NOT the `xoxb-…` bot token.

Fix. Confirm `HERALD_SLACK_APP_TOKEN` starts with `xapp-`. If not, regenerate it from **Basic Information → App-Level Tokens** with the `connections:write` scope.

§9.5 Attachments arriving in the wrong inbox subdir

Symptom. A Slack photo lands under `~/.herald/inbox/tgram/` (or vice-versa).

Root cause. Either `HERALD_INBOX_DIR` is set to a path that overlaps another deployment, OR the adapter's `Name()` does not match its registered name (a refactor regression).

Fix.

1. Check `HERALD_INBOX_DIR` for an unintended override.
2. Confirm the adapter's `Name()` method returns the same string passed to `channels.Register()` at `init()` time — they MUST match or the inbox helper picks the wrong subdir.

3. Grep your code for `channels.WriteContentAddressed("<literal-string>", …)` — a hard-coded literal that drifts from `Name()` is a §107 hazard.

§9.6 `HERALD_CHANNELS=foo` failing at boot

Symptom. `pherald listen: build "foo" channel: channels: unknown channel: "foo" (registered: [slack tgram]).`

Root cause. Typo, or the adapter's package is not blank-imported by `pherald/cmd/pherald/listen.go` (so its `init()` never ran).

Fix.

1. Check spelling — compare against `channels.Names()` output (embedded in the error).
2. Confirm the blank import: `grep -E '_ "github.com/vasic-digital/herald/commons_messaging/channels/' pherald/cmd/pherald/listen.go`. Every channel you expect to be available MUST have a matching blank import.

§9.7 Privacy-mode Telegram: bot sees no group messages

Symptom. Telegram DMs work; group messages never arrive.

Root cause. Telegram bots default to privacy-mode-on, which means they only see commands (`/foo`) and replies + mentions in groups.

Fix. DM `@BotFather`, `/setprivacy`, select your bot, choose `Disable`. Restart `pherald listen`. The bot will now see every message in groups it is a member of.

§9.8 Telegram `Conflict: terminated by other getUpdates request`

Symptom. Long-poll boots, then aborts with `Conflict`.

Root cause. Another process is calling `getUpdates` against the same token — Telegram permits only one consumer per token. Usually a second `pherald listen` instance, a forgotten dev shell, or a webhook configured against the same token.

Fix. Kill the competing consumer (`pkill -f 'pherald listen'`). If a webhook is the culprit, delete it: `curl -X POST https://api.telegram.org/bot<TOKEN>/deleteWebhook`.

§9.9 Inbox files are zero-byte / `.part` residue

Symptom. `~/.herald/inbox/<channel>/dl-*.part` files left behind, or content-addressed final files are zero-byte.

Root cause. A streaming download was interrupted (network drop, process kill) before the atomic rename completed; OR a bug in a new adapter's `DownloadAttachment` returned success before fully writing.

Fix.

1. Delete the `.part` files — they are deliberate temp residue, the next download will start fresh and idempotently land the final file.
2. If zero-byte FINAL files appear, the new adapter has a bug. Re-check that it streams the body through `io.MultiWriter(file, hasher)` and ONLY returns the path after `os.Rename` succeeds.

§9.10 Cross-channel "wrong reply destination"

Symptom. A reply intended for Slack arrives in Telegram (or vice-versa).

Root cause. The dispatcher's `channelRouter` keys replies by `recipient.Channel` — if an adapter constructed a `commons.Recipient` with the wrong `Channel` value, the router sends the reply via the wrong adapter.

Fix. In your adapter's Subscribe handler, confirm every `commons.Recipient` is constructed with `Channel: a.Name()` (the adapter's own name). The `channelRouter` test fixture (`pherald/cmd/pherald/listen_test.go`) pins this for the two live channels; a third adapter must add a parallel assertion.

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§10. Audit checklist (operator pre-deploy)

Run through the checklist before every fresh deploy or after every credential rotation.

§10.1 Environment

- [ ] Determine the intended channel set; export `HERALD_CHANNELS=<comma-list>` (or leave unset for Telegram-only).
- [ ] For every channel in the set, every required env var listed in the channel's section is exported.
- [ ] Token prefixes match the channel's expected format:
  - [ ] Telegram token: numeric id, colon, alphanumeric — e.g. `1234567890:AAH…`.
  - [ ] Slack bot token: starts `xoxb-`.
  - [ ] Slack app token: starts `xapp-` (only if Slack is in `HERALD_CHANNELS`).
- [ ] Slack channel id: starts `C` (public) or `G` (private), NOT a `#name`.
- [ ] Telegram chat id: numeric (negative for groups, `-100…` for supergroups, positive for DMs).
- [ ] No credential appears in `git ls-files | xargs grep -l '<your-token>'` (manual spot-check before commit).

§10.2 Bot setup

- [ ] **Telegram** — privacy-mode disabled (`/setprivacy` → `Disable` in BotFather) IF the bot is meant to see group messages.
- [ ] **Telegram** — no other consumer is calling `getUpdates` against the same token. `curl -s 'https://api.telegram.org/bot<TOKEN>/getWebhookInfo' | jq` confirms no stale webhook is configured.
- [ ] **Slack** — bot is a member of the target channel: `/invite @your-bot-name` in Slack.
- [ ] **Slack** — Socket Mode is enabled in **App settings → Socket Mode**.
- [ ] **Slack** — Bot Token Scopes include all of: `chat:write`, `files:read`, `files:write`, `channels:history` (and/or `groups:history`).
- [ ] **Slack** — App-Level Token scope: `connections:write`.

§10.3 Registry resolution

- [ ] Confirm every enabled channel is registered. From the repo root: `grep 'channels.Register' commons_messaging/channels/*/*.go` lists every available channel. The set MUST contain every name listed in `HERALD_CHANNELS`.
- [ ] (Future) A `pherald listen --dry-run` (planned — file as a follow-up HRD if needed) prints `channels.Names()` + the resolved per-channel config without crossing the wire.

§10.4 Filesystem

- [ ] `~/.herald/inbox/` exists with mode `0700` (or `$HERALD_INBOX_DIR`, if overridden).
- [ ] Per-channel subdirs are present for every enabled channel after the first inbound attachment arrives — `ls -la ~/.herald/inbox/`.
- [ ] Disk has headroom for the worst-case attachment burst (`AttachmentMaxMiB` × expected concurrent uploads).

§10.5 §107.x evidence

- [ ] Plan the run-id directory for this session: `docs/qa/HRD-<NNN>-<TS>Z/`.
- [ ] If using qaherald, point its `--qa-out-dir` at that path and confirm a transcript / attachments tree is captured before tagging.

- [ ] For Telegram-driven manual runs, screenshot both directions of the conversation and commit the screenshots in the run-id directory.
- [ ] For Slack-driven manual runs, save the conversation transcript (Slack right-click → **Copy link**, or the App Home transcript export) and commit it.

§10.6 First-message smoke

- [ ] Start `pherald listen` in the foreground; observe the boot lines for each enabled channel:
  - [ ] `pherald listen: starting Telegram getUpdates long-poll loop`
  - [ ] (Slack equivalent — Socket Mode WebSocket connected).
- [ ] Send a test message from a human user. Confirm:
  - [ ] The bot replies (Claude Code dispatch round-trip).
  - [ ] The reply quotes the original message (`reply_to_message_id` / `thread_ts`).
  - [ ] The conversation appears in `docs/herald/diary/main.md`.
- [ ] Send a test message FROM the bot to itself (loopback). Confirm the self-filter drops the echo — no second reply arrives, no Claude Code dispatch in the logs.

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§11. References

§11.1 Source files

- `commons_messaging/channels/channel.go` — the eight-method interface + `SelfIdentity` types.
- `commons_messaging/channels/registry.go` — `Register` / `New` / `Names` + `ErrUnknownChannel`.
- `commons_messaging/channels/inbox.go` — `InboxDir`, `WriteContentAddressed`, `MimeToExt`.
- `commons_messaging/channels/selffilter.go` — `StampSender`, `IsSelfEcho`.
- `commons_messaging/channels/tgram/*` — Telegram adapter (Wave 6 substrate).
- `commons_messaging/channels/slack/*` — Slack adapter (Wave 7 T6 reference implementation).
- `pherald/cmd/pherald/listen.go` — `loadEnabledChannels`, `perChannelConfig`, `channelRouter`, `runListen`.

§11.2 Spec V3

- §11 — Channel capabilities matrix (every channel's row).
- §11.0 — outbound `commons.Channel` contract (the five embedded methods).
- §32.2 — inbound long-poll / Socket Mode requirements.
- §32.9 — anti-echo-loop self-filter mandate.
- §43 — Channel + flavor catalogue.
- §107.x — `docs/qa/<run-id>/` evidence mandate (Helix §11.4.83 cascade).

§11.3 HRDs

HRD	Wave	Task	Description
HRD-110	7	T1	Extract `channels.Channel` interface (richer inbound contract embedding §11.0).
HRD-111	7	T2	`init()`-based channel registry + `channels.New` / `channels.Names`.
HRD-112	7	T3	Per-channel inbox subdirs (`~/.herald/inbox/<channel>/`).
HRD-113	7	T4	Generalize bot self-filter via `BotSelfIdentity` + `IsSelfEcho`.
HRD-114	7	T5	Multi-channel `pherald listen` (fan-in + `channelRouter`).
HRD-115	7	T6	Slack channel adapter (Socket Mode + Web API).
HRD-116	7	T7	qaherald Slack `MessengerClient` + scenario builder.
HRD-117	7	T8	Spec V3 §11.0 + §32.2 + §43 update.
HRD-118	7	T9	`scripts/e2e_bluff_hunt.sh` multi-channel invariants (E81..E88).
HRD-119	7	T10	Wave 7 paired §1.1 mutation gate (`tests/test_wave7_mutation_meta.sh`).
HRD-120	7	T11	§11.4.85 stress + chaos for the multi-channel runtime.
HRD-121	7	T12	This guide + Issues→Fixed migration + v0.6.0 release prep.

§11.4 Related guides

- `OPERATOR_CREDENTIALS.md` — umbrella credentials guide; defines the `.env` resolution order and the dual-source model (shell exports vs `.env`).
- `messengers/TELEGRAM.md` — Telegram-specific deep-dive (HRD-011).
- `messengers/SLACK.md` — Slack-specific deep-dive (HRD-115; landing alongside this guide).
- `HERALD_CONSTITUTION.md` — §107 end-user-usability covenant + §107.x evidence mandate.
- `CONSTITUTION_INHERITANCE.md` — parent-discovery rules.

§11.5 Wave 7 plan

- `docs/superpowers/plans/2026-05-27-wave7-generic-messenger.md` — the full Wave 7 plan (Tasks 1..12, locked decisions, file structure).

§11.6 Constitutional anchors

- Helix §11.0 — outbound `Channel` contract.
- Helix §11.4 / §11.4.5 / §11.4.6 — anti-bluff captured-evidence mandate.
- Helix §11.4.83 — `docs/qa/<run-id>/` evidence cascade (Herald §107.x).

- Helix §11.4.85 — stress + chaos test mandate (T11 binding).
- Helix §11.4.94 — zero-idle parallel-by-default operating mode.
- Helix §11.4.96 — safe-parallel-work catalogue.

Sources verified

Per HelixConstitution §11.4.99 + Herald §108.n (Latest-Source Documentation Cross-Reference Mandate). Every operator-facing instruction in this document was cross-referenced against the LATEST official online documentation of the relevant service before publication.

Last verified: 2026-05-28

Source	URL / path	Authored / verified
Telegram official Bot API documentation	https://core.telegram.org/bots/api	§3 (`HERALD_TGRAM_BOT_TOKEN` format from BotFather; `HERALD_TGRAM_CHAT_ID` per-chat-type shape — private positive / group negative / supergroup `-100…`); §3.2 (`/newbot` username-ending-in-bot rule, `/setprivacy` Disable flow); §3.3 (chat-id discovery via `getUpdates`); §3.4 (sendMessage / sendPhoto / sendDocument / sendVoice / MarkdownV2 wire); §3.4 (long-poll 25 s timeout + `getUpdates` semantics); §3.5 (`AttachmentMaxMiB = 50` hard cap); §9.7 + §9.8 (privacy-mode, `Conflict: terminated by other getUpdates request`).
Telegram Bot Features — Privacy Mode	https://core.telegram.org/bots/features#privacy-mode	§3.2 step 4 (`/setprivacy` Disable); §9.7 (the "privacy-mode-on is the #1 stuck-cause" guidance).
Slack Web API method index	https://api.slack.com/methods	§4 (Slack adapter overview — `chat.postMessage`, `files.uploadV2`, `auth.test`, `conversations.history`); §4.5 (xoxb- token redaction rationale — workspace-scope credential semantics); §9.2 (`channel_not_found`); §9.3 (`invalid_auth` / `token_expired`).
Slack Events API + Socket Mode	https://api.slack.com/apis/connections/socket + https://api.slack.com/apis/events-api	§4.2 (Socket Mode WebSocket setup — `connections:write` scope on app-level token; `xapp-` prefix; `Enable Socket Mode` toggle; `apps.connections.open` dial); §4.3 (Socket Mode == no inbound NAT traversal / no public URL); §9.4 (`not_allowed_token_type` for `apps.connections.open`).
Slack OAuth + Bot Token Scopes	https://api.slack.com/scopes	§4.2 step 3 (`chat:write` / `files:read` / `files:write` /

Source	URL / path	Authored / verified
		`channels:history` / `groups:history` / `users:read` Bot Token Scopes); §4.2 step 6 (`xoxb-` Bot User OAuth Token install); §10.2 audit-checklist entries.
Slack reply threading (`thread_ts`)	https://api.slack.com/messaging/retrieving#threads	§4.3 (thread_ts as Slack's reply anchor; per-channel router's reply-routing key).
telebot.v3 vendored library	`submodules/telebot/` (v3.3.8, pinned per §11.4.74)	§1.4 (`init()`-registered `tgram` constructor); §6.3 (`bot.Me.Username` → `IdentityUsername`).
slack-go vendored library	`submodules/slack-go/`	§4 + §9 (Slack adapter behaviour — Web API client, Socket Mode handler, retry semantics).
Empirical Herald operator testing 2026-05-28	`docs/qa/HRD-LIVE-20260528T082128Z/`	§3 LIVE-since-Wave-6 status; §4 Slack LIVE-since-Wave-7-T6 status; §1.5 multi-channel fan-in semantics.
HelixConstitution §11.4.99 (this document's authority)	`<parent>/constitution/Constitution.md` §11.4.99 (HelixConstitution commit `c640947`)	This footer pattern + 90-day cadence.

Re-verification cadence (per §11.4.99 (C)): Both Telegram and Slack are risk-classified services (per §11.4.99 (D) — bot APIs face anti-abuse / token-revocation; OAuth scope changes can break installed apps silently). Telegram-specific subsections (§3, §9.7, §9.8) → **90-day max staleness**, next due **2026-08-26**. Slack-specific subsections (§4, §9.2..§9.4) → **90-day max staleness**, next due **2026-08-26**. Channel-agnostic substrate (§1, §2, §5, §6, §7, §8, §10) → **180-day max staleness**, next due **2026-11-24**. Re-verify earlier on: Telegram Bot API changelog entry, Slack API changelog entry (https://api.slack.com/changelog), operator-error reports, Herald vN.0.0 release boundary.

End of guide.